

Model Evaluation & Refinement Report

1. Executive Project Summary

This comprehensive technical evaluation reports on the end-to-end training, validation, and architectural refinement phases conducted across three separate engineering paradigms: Tabular Machine Learning (XGBoost), Deep Learning Sequence Networks (LSTM), and Classical Statistical Modeling (SARIMAX). The core mission was to construct a robust predictive system capable of tracking traffic volume variations across complex urban networks.

2. Paradigm Leaderboard & Error Analysis

To evaluate model accuracy and structural generalizability cleanly across chronological horizons without future leakage risk, a strict **Time Series Cross-Validation Split Loop** was executed uniformly. Performance was cataloged using Mean Absolute Error (MAE) and Score Variance (Standard Deviation).

Model Framework Paradigm	Architectural Configuration Details	Validation MAE	Score Variance (σ)
Tuned XGBoost Regressor	Shallow Depth (max_depth=4) + Custom Interaction Flags	13.82 vehicles	3.18 vehicles
Refined LSTM Network	50 Memory Channels + 0.4 Dropout Layer + L2 Regularization	14.67 vehicles	2.78 vehicles
Classical SARIMAX Model	Order (2,0,0) x (1,0,0) ²⁴ + External Exogenous Channels	27.52 vehicles	4.92 vehicles

3. XGBoost Architecture & Feature Engineering

The baseline tree model initially demonstrated severe systemic under-guessing bugs specifically tracking the 08:00 AM weekday commuter surge. Diagnostic residual checks exposed that the tree splitter evaluated chronological columns independently, missing interactive non-linear trends.

This was resolved by implementing targeted **Interaction Features** in the data preprocessing matrix, defining explicit morning and evening peak traffic flags:

$$is_weekday_morning_rush = (hour_only \geq 7) \times (hour_only \leq 9) \times (is_weekend == 0)$$

Hyperparameter optimization using a randomized grid regularized the trees, lowering max depth from 6 to 4 and learning rate to 0.01. This successfully prevented the model from overfitting to short-term data drift, achieving the project-high accuracy of **13.82 MAE**.

4. Deep Learning Sequence Modeling (LSTM)

To evaluate sequence-aware networks, tabular data was reshaped into a continuous 3D tensor array with dimensions matching:

$$\text{Shape Tensor} = (\text{Samples}, \text{Time Steps [24]}, \text{Features})$$

The baseline LSTM network achieved a training error of 12.21 but a validation error of 19.02, signaling severe high variance. To force structural generalization, the architecture was injected with an escalated 0.4 Dropout block and Kernel L2 weight penalties. This stabilized the model, delivering the lowest standard deviation across time horizons (**2.78 σ**), proving that sequence tracking networks handle macro-seasonal drift with maximum consistency.

5. Classical Statistical Modeling (SARIMAX)

Stationarity diagnostics were verified via the Augmented Dickey-Fuller (ADF) test, yielding an ADF statistic of -15.57 and a definitive p-value of 0.0000. This statistical profile permitted an integration parameter of **$d = 0$** . Parameter identification via Autocorrelation (ACF) and Partial Autocorrelation (PACF) plots displayed clear local cutoff thresholds at Lag 2 and prominent daily cycles at Lag 24.

While univariate modeling failed catastrophically due to its blindness to calendar horizons (31.39 MAE), adding exogenous variables (SARIMAX) lowered the error to 27.52. The performance ceiling highlights the "Linearity Trap" of classical statistics; it applies rigid offsets and fails to navigate interactive non-linear cycle path transformations.

6. Production Deployment Strategy

Based on verified validation data matrices, the **Tuned XGBoost Regressor** is designated as the primary production model engine due to its superior capacity to execute crisp boundary splits on the engineered rush hour interactions. The LSTM architecture is recommended as a secondary validator to track long-term baseline drift across multi-month horizons.